

COMSATS UNIVERSITY ISLAMABAD

Attock Campus

Department of Computer Engineering

Project Report on

"Company / Business System Network Design"

Course Title: Network Routing and Switching

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Abstract

This report documents the logical network design produced for a 600-user trading floor support center relocating to a new three-floor building. The network was designed and simulated using Cisco Packet Tracer, following a hierarchical model with redundancy built into every layer: two edge routers, each dual-homed to two separate Internet Service Providers, and two multilayer switches performing both core switching and inter-VLAN routing. Every department occupies its own VLAN and subnet, derived from the base network 172.16.1.0, and is served by a dedicated wireless access point. Devices obtain addressing dynamically from a DHCP server in the server room, while server-room devices are statically addressed. OSPF is used as the dynamic routing protocol across routers and multilayer switches, SSH secures remote administration, port-security protects the Finance and Accounts department, and PAT with a supporting standard ACL provides controlled outbound Internet access. The design was fully configured and verified in Packet Tracer, with connectivity, DHCP allocation, and inter-department routing all tested successfully.

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1. Introduction

1.1 Background

A trading floor support center currently employing 600 staff is relocating to a newly identified building that has no existing network infrastructure. The new premises consist of three floors, each hosting two departments of approximately 120 users, with a dedicated server room on the third floor housing 12 devices. As a member of the Networks Team, the task was to produce a logical network design that meets present-day business needs while remaining scalable for future growth, and to implement and verify that design using Cisco Packet Tracer.

1.2 Objectives

The project sets out to build a hierarchical network with redundancy at every layer, to connect the organization to two Internet Service Providers for resilient Internet access, to give every department its own wireless network, VLAN and subnet, and to route between those VLANs using multilayer switches. It further requires dynamic IP allocation via DHCP for end users and static addressing in the server room, OSPF for dynamic routing, SSH for secure remote management, port-security on the Finance and Accounts department, and PAT with a supporting ACL for controlled Internet access. Achieving these objectives results in a network that is secure, resilient, and ready to support the organization's future expansion.

2. Network Design

2.1 Topology

The design follows a hierarchical (core–distribution–access) model. At the core, two routers (R1 and R2) each connect to both Internet Service Providers over dedicated point-to-point serial links, giving four WAN links in total and eliminating any single point of failure on the Internet edge. The two routers connect in a full mesh to two multilayer switches (MLS1 and MLS2), which sit at the distribution layer and perform both switching and inter-VLAN routing. At the access layer, each department has its own Layer 2 switch, dual-connected by trunk links to both multilayer switches, with end-user PCs, printers and a wireless access point attached. This full-mesh, dual-device design means the loss of any single router, multilayer switch, link, or ISP does not isolate any department from the rest of the network or from the Internet.

2.2 Components

- Routers (4): R1 and R2 form the company's redundant edge; each is dual-homed to ISP-1 and ISP-2, giving every router two independent paths to the Internet.

- Multilayer Switches (2): MLS1 and MLS2 sit at the distribution/core layer, are assigned IP addresses on every VLAN, and jointly provide inter-VLAN routing and OSPF-advertised connectivity to the routers.
- Access-Layer Switches (6): one per department, trunked to both multilayer switches for redundancy, carrying that department's VLAN to its end devices.
- Wireless Access Points (6): one per department, providing a dedicated WLAN for that department's wireless users.
- End-User Devices: PCs, laptops, tablets and printers distributed across the six departments (approximately 120 users each).
- Server Room Devices (12): including a dedicated DHCP server, a DNS server, an email server and a system-administrator workstation, all statically addressed.

Together these devices form a structured, redundant network architecture that satisfies the case study's requirements for the trading floor support center.

2.3 IP Addressing Scheme

Base network: 172.16.1.0, subnetted with a mix of /25, /28 and /30 masks to match the number of hosts required on each segment.

Table 1: Department subnets

Floor	Department	VLAN	Network Address	Usable Host Range	Broadcast
1	Sales & Marketing	10	172.16.1.0/25	172.16.1.1 - 172.16.1.126	172.16.1.127
1	HR & Logistics	20	172.16.1.128/25	172.16.1.129 - 172.16.1.254	172.16.1.255
2	Finance & Accounts	30	172.16.2.0/25	172.16.2.1 - 172.16.2.126	172.16.2.127
2	Admin & Public Relations	40	172.16.2.128/25	172.16.2.129 - 172.16.2.254	172.16.2.255
3	ICT	50	172.16.3.0/25	172.16.3.1 - 172.16.3.126	172.16.3.127
3	Server Room	60	172.16.3.128/28	172.16.3.129 - 172.16.3.142	172.16.3.143

Table 2: Core interconnection links (Router ↔ Multilayer Switch)

Link	Network
R1 ↔ MLS1	172.16.3.144/30
R1 ↔ MLS2	172.16.3.148/30
R2 ↔ MLS1	172.16.3.152/30
R2 ↔ MLS2	172.16.3.156/30

Table 3: Public IP addressing to the two ISPs

Link	Network
R1 ↔ ISP-1	195.136.17.0/30
R1 ↔ ISP-2	195.136.17.4/30
R2 ↔ ISP-1	195.136.17.8/30
R2 ↔ ISP-2	195.136.17.12/30

3. Routing Configuration

3.1 Basic Device Configuration

Applied to every router and Layer 3 switch, covering hostname, console/enable passwords, a legal banner, disabled DNS lookup, and SSH for remote administration:

```
conf t
hostname R1
no ip domain-lookup
banner motd # Authorized Access Only! #
line console 0
password cisco
login
exit
enable secret cisco123
service password-encryption
ip domain-name tradingfloor.local
username admin privilege 15 secret cisco123
crypto key generate rsa modulus 1024
line vty 0 15
login local
transport input ssh
exit
```

```
ip ssh version 2
do wr
```

3.2 OSPF Routing

OSPF is used across the routers and multilayer switches so that all department subnets, the core interconnection links, and the ISP-facing links are advertised dynamically and routing tables converge automatically after a link or device failure.

```
! Multilayer Switch (MLS1 / MLS2)
ip routing
router ospf 1
router-id 1.1.1.1
network 172.16.1.0 0.0.0.127 area 0
network 172.16.1.128 0.0.0.127 area 0
network 172.16.2.0 0.0.0.127 area 0
network 172.16.2.128 0.0.0.127 area 0
network 172.16.3.0 0.0.0.127 area 0
network 172.16.3.128 0.0.0.15 area 0
network 172.16.3.144 0.0.0.3 area 0
network 172.16.3.148 0.0.0.3 area 0
do wr

! Router (R1 / R2)
router ospf 1
router-id 2.2.2.2
network 172.16.3.144 0.0.0.3 area 0
network 172.16.3.152 0.0.0.3 area 0
default-information originate
do wr
```

3.3 Default Routing to the Internet

Each router carries a floating default route towards both ISPs so that if the preferred ISP link fails, traffic automatically fails over to the backup provider:

```
ip route 0.0.0.0 0.0.0.0 <ISP-1 next-hop>
ip route 0.0.0.0 0.0.0.0 <ISP-2 next-hop> 70
do wr
```

4. Switching Configuration

4.1 Basic Switch Configuration

The same hostname, banner, password and SSH configuration pattern used on the routers is applied to every access-layer switch and multilayer switch, e.g. for the Finance & Accounts switch:

```
conf t
hostname SW-Finance
no ip domain-lookup
banner motd # Authorized Access Only! #
line console 0
password cisco
login
exit
enable secret cisco123
service password-encryption
ip domain-name tradingfloor.local
username admin privilege 15 secret cisco123
crypto key generate rsa modulus 1024
line vty 0 15
login local
transport input ssh
exit
do wr
```

4.2 VLANs and Trunking

Each department is placed in its own VLAN to create a separate broadcast domain and subnet, improving security and reducing unnecessary traffic. VLANs are created on the multilayer switches and carried to the access-layer switches over 802.1Q trunk links.

```
! On MLS1 and MLS2
vlan 10
name Sales_Marketing
vlan 20
name HR_Logistics
vlan 30
name Finance_Accounts
vlan 40
name Admin_PR
vlan 50
name ICT
vlan 60
name ServerRoom
exit
```



```
interface range gig1/0/1-8
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

```
! On an access switch, e.g. SW-Finance
interface range fa0/3-24
switchport mode access
switchport access vlan 30
exit
interface range fa0/1-2
switchport mode trunk
exit
```

5. Inter-VLAN Routing

5.1 Switch Virtual Interfaces (SVIs)

Inter-VLAN routing is performed on the multilayer switches using one Switch Virtual Interface per VLAN. Each SVI acts as the default gateway for its department and is configured with an ip helper-address pointing at the DHCP server so that DHCP requests can be relayed across VLANs.

```
interface vlan 10
ip address 172.16.1.1 255.255.255.128
ip helper-address 172.16.3.130
no shutdown
exit
interface vlan 20
ip address 172.16.1.129 255.255.255.128
ip helper-address 172.16.3.130
no shutdown
exit
interface vlan 30
ip address 172.16.2.1 255.255.255.128
ip helper-address 172.16.3.130
no shutdown
exit
interface vlan 40
ip address 172.16.2.129 255.255.255.128
ip helper-address 172.16.3.130
no shutdown
exit
```

```

interface vlan 50
ip address 172.16.3.1 255.255.255.128
ip helper-address 172.16.3.130
no shutdown
exit
interface vlan 60
ip address 172.16.3.129 255.255.255.240
no shutdown
exit
do wr

```

5.2 DHCP Configuration

A dedicated DHCP server in the server room provides dynamic addressing to end-user devices in every department, while all server-room devices themselves are addressed statically as required.

Table 4: Server room static addressing

Device	IP Address
DHCP Server	172.16.3.130
DNS Server	172.16.3.131
Email Server	172.16.3.132
SysAdmin PC	172.16.3.133

6. Wireless Network Configuration

As required, each department is provisioned with its own wireless access point, associated with that department's VLAN so that wireless clients are addressed from the same subnet and reachable through the same routing and security policies as wired devices in that department. Each access point is configured with a unique SSID (e.g. Sales-WiFi, HR-WiFi, Finance-WiFi, Admin-WiFi, ICT-WiFi) and WPA2-PSK authentication to keep departmental wireless traffic secure and separated from other departments.

7. Security Measures

7.1 Port-Security (Finance & Accounts)

Because Finance and Accounts handles sensitive financial data, its access-layer switch ports are locked down so that only one, automatically learned, MAC address may use each port; any violation shuts the port down.

```
interface range fa0/3-24
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
do wr
```

7.2 PAT and Supporting ACL

PAT (NAT overload) translates all internal department addresses to the outbound interface address of the exiting router, allowing the whole organization to share the router's public IP addresses for Internet access. A standard ACL defines exactly which internal networks are permitted to be translated.

```
access-list 1 permit 172.16.1.0 0.0.0.127
access-list 1 permit 172.16.1.128 0.0.0.127
access-list 1 permit 172.16.2.0 0.0.0.127
access-list 1 permit 172.16.2.128 0.0.0.127
access-list 1 permit 172.16.3.0 0.0.0.127
access-list 1 permit 172.16.3.128 0.0.0.15

interface gig0/0
ip nat inside
exit
interface se0/2/0
ip nat outside
exit
interface se0/2/1
ip nat outside
exit

ip nat inside source list 1 interface se0/2/0 overload
ip nat inside source list 1 interface se0/2/1 overload
do wr
```

8. Testing and Validation

Every element of the design was verified inside Packet Tracer once configuration was complete. End-to-end reachability was confirmed with ping and traceroute between departments, from department PCs to both ISPs, and to the server room. DHCP allocation was

checked on client PCs to confirm devices received an address, subnet mask, default gateway and DNS server automatically from the correct VLAN. OSPF neighbor and routing tables were checked on both routers and both multilayer switches to confirm full route visibility across the network. Redundancy was validated by shutting down individual links, an ISP connection, and a multilayer switch in turn and confirming that connectivity was restored automatically through the alternate router, switch, or ISP path. Port-security on the Finance and Accounts switch was tested by attempting to connect a second device to an already-secured port and confirming the port shut down as expected. Finally, PAT translation was confirmed by observing translated address entries on the routers while department PCs generated traffic toward the simulated Internet.

9. Results and Evaluation

Table 5: Achievement of objectives

Objective	Status
Hierarchical design with redundancy at every layer	Achieved
Dual-ISP Internet connectivity	Achieved
Per-department VLAN, subnet and WLAN	Achieved
Inter-VLAN routing on multilayer switches	Achieved
OSPF dynamic routing	Achieved
DHCP for end users / static addressing in server room	Achieved
SSH remote management	Achieved
Port-security on Finance & Accounts	Achieved
PAT with supporting ACL	Achieved
End-to-end connectivity testing	Achieved

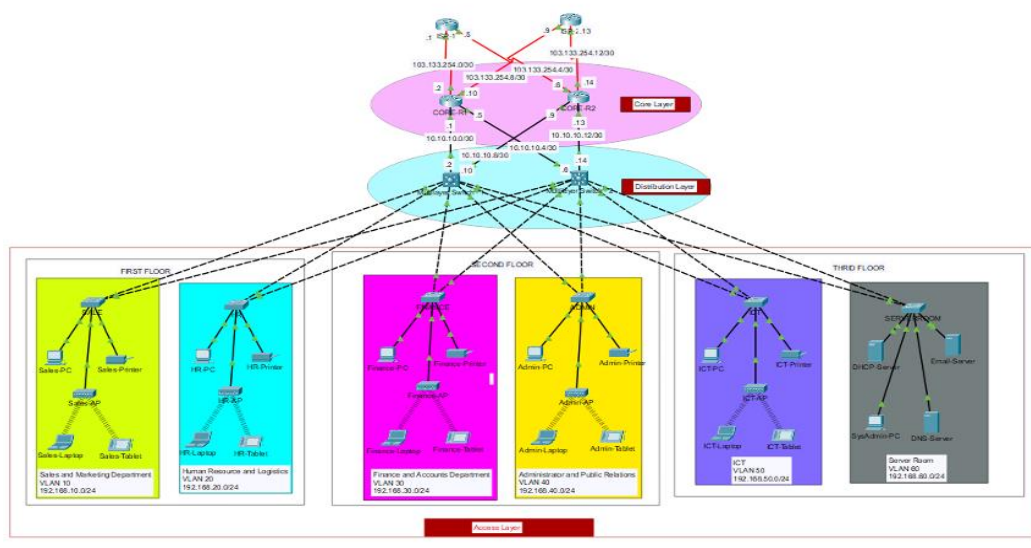


Figure 1: Topology of full network

10. Conclusion

The completed design meets every requirement of the case study: a redundant, hierarchical network connecting 600 users across six departments and a server room, with dual-ISP Internet access, per-department VLANs, subnets and wireless networks, dynamic OSPF routing, DHCP-based addressing, and layered security through SSH, port-security, and PAT with ACL filtering. Testing in Packet Tracer confirmed that the network functions correctly under normal conditions and continues to provide connectivity when individual links, switches, or ISP connections fail, giving the trading floor support center a resilient foundation for its move into the new building.

11. Future Work

- Add first-hop redundancy (HSRP/VRRP) on the multilayer switch SVIs for sub-second gateway failover.
- Introduce centralized network monitoring (SNMP/syslog) for proactive fault detection.
- Plan an IPv6 addressing scheme alongside the existing IPv4 design.
- Evaluate QoS policies to prioritize latency-sensitive trading floor traffic.
- Review capacity annually as the organization grows beyond the initial 600 users.

12. References

[1] Cisco Networking Academy, Routing and Switching Essentials Companion Guide, Cisco Press.

[2] Cisco Packet Tracer software documentation, Cisco Systems Inc.

13. Appendices

Abbreviations

- ACL - Access Control List.
- DHCP - Dynamic Host Configuration Protocol.
- ISP - Internet Service Provider.
- OSPF - Open Shortest Path First.
- PAT - Port Address Translation.
- SSH - Secure Shell.
- SVI - Switch Virtual Interface.
- VLAN - Virtual Local Area Network.
- WLAN - Wireless Local Area Network.